In [2]:
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# 🏠 House Price Prediction using Machine Learning
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import joblib


# 🔹 Step 2: Create & Load Dataset
data = {
    'Area': [2600, 3000, 3200, 1800, 2200, 4000, 1500, 2700, 3500, 4200],
    'Bedrooms': [4, 3, 4, 2, 3, 5, 2, 3, 4, 5],
    'Bathrooms': [3, 2, 3, 1, 2, 4, 1, 2, 3, 4],
    'Stories': [2, 1, 2, 1, 2, 2, 1, 1, 2, 3],
    'Parking': [1, 2, 1, 0, 1, 2, 1, 1, 2, 2],
    'Price': [55000000, 45000000, 60000000, 30000000, 40000000, 85000000, 2500
}

df = pd.DataFrame(data)
df.to_csv("house_data.csv", index=False)
print("✅ Dataset created successfully!\n")
display(df.head())


# 🔹 Step 3: Data Overview
print("\n📊 Dataset Info:")
print(df.info())

print("\n📈 Statistical Summary:")
display(df.describe())


# 🔹 Step 4: Data Visualization
plt.figure(figsize=(7,5))
sns.heatmap(df.corr(), annot=True, cmap="YlGnBu")
plt.title("Correlation Heatmap")
plt.show()

plt.figure(figsize=(7,5))
sns.pairplot(df)
plt.suptitle("Feature Relationships", y=1.02)
plt.show()


# 🔹 Step 5: Data Preparation
X = df[['Area', 'Bedrooms', 'Bathrooms', 'Stories', 'Parking']]
y = df['Price']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, rando
print("✅ Data split into training and testing sets successfully!")


# 🔹 Step 6: Train the Model
model = RandomForestRegressor(n_estimators=100, random_state=42)
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model.fit(X_train, y_train)
print("✅ Model training completed!")


#  🔹 Step 7: Evaluate Model
y_pred = model.predict(X_test)

r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print("\n📊 Model Evaluation Results:")
print(f"R2 Score: {r2:.3f}")
print(f"Mean Absolute Error: {mae:.2f}")
print(f"Root Mean Squared Error: {rmse:.2f}")


#  🔹 Step 8: Visualization - Actual vs Predicted
plt.figure(figsize=(7,5))
plt.scatter(y_test, y_pred, color='blue', label="Predicted vs Actual")
plt.plot([min(y_test), max(y_test)], [min(y_test), max(y_test)], color='red',
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.legend()
plt.show()


#  🔹 Step 9: Prediction Function
def predict_price(area, bedrooms, bathrooms, stories, parking):
    features = np.array([[area, bedrooms, bathrooms, stories, parking]])
    price = model.predict(features)[0]
    return round(price, 2)

# Example prediction
example_price = predict_price(3000, 3, 2, 1, 2)
print(f"\n💰 Predicted Price for [3000 sq.ft, 3BHK, 2 bath, 1 story, 2 parking


#  🔹 Step 10: Save and Reload Model
joblib.dump(model, "house_price_model.pkl")
print("\n✅ Model saved as 'house_price_model.pkl'")

loaded_model = joblib.load("house_price_model.pkl")
print("♻️ Model reloaded successfully!")

# Test reloaded model
test_price = loaded_model.predict(np.array([[2500, 3, 2, 1, 1]]))[0]
print(f"🏠 Reloaded Model Prediction (2500 sq.ft): ₹{test_price:,.0f}")


#  🔹 Step 11: Graph - Price vs Area
plt.figure(figsize=(8,5))
sns.scatterplot(x='Area', y='Price', data=df, s=100, color='green')
plt.title("Price vs Area Relationship")
plt.xlabel("Area (sq.ft)")
plt.ylabel("Price (₹)")
```

```
plt.show()
import warnings
warnings.filterwarnings('ignore')

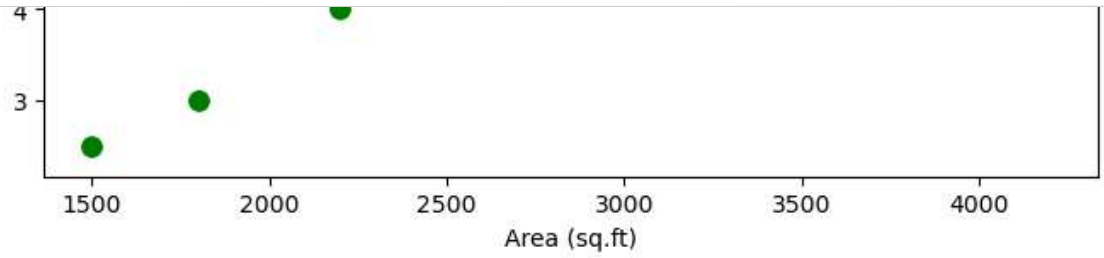
print("\n🏠 House Price Prediction System 🧠")

# Take input from user
area = float(input("Enter area (in sq.ft): "))
bedrooms = int(input("Enter number of bedrooms: "))
bathrooms = int(input("Enter number of bathrooms: "))
stories = int(input("Enter number of stories: "))
parking = int(input("Enter number of parking spaces: "))

# Predict using trained model
predicted_price = model.predict([[area, bedrooms, bathrooms, stories, parking]

print(f"\n💰 Estimated House Price: ₹{predicted_price:,.0f}")

print("\n🎯 Project Completed Successfully!")
```

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🏠 House Price Prediction System 🧠
Enter area (in sq.ft): 1000
Enter number of bedrooms: 3
Enter number of bathrooms: 2
Enter number of stories: 1
Enter number of parking spaces: 2

💰 Estimated House Price: ₹43,690,000

🎯 Project Completed Successfully!
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